LAB 6: CODE AND BRANCH COVERAGE ANALYSIS

Self Storage Application — Test Coverage Report

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Repository	github.com/aryaappaji25/L6_L7_51_self_storage

1. Objective

The objective of this lab is to create and execute tests for the Self Storage Python application, measure code coverage and branch coverage, and document the results. The tests exercise the application's storage-rent calculation, access-control decisions, and payment-status logic.

2. Tools and Files

Item	Purpose
Python	Runs the application and test code.
pytest	Discovers and executes automated tests.
coverage.py	Measures statement coverage and branch coverage.
self_storage.py	Application module under test.
CodeCoverageTest_51_self_storage.py	Tests intended to measure code coverage.
BranchCoverageTest_51_self_storage.py	Tests intended to exercise decision branches.

3. Testing Methodology

Tests were written with pytest and run separately for the code-coverage and branch-coverage test files. The coverage tool was run with the application module as the measurement source. For branch coverage, the branch option was enabled. The test cases cover representative inputs and outcomes, including valid and invalid unit sizes, month values, membership and climate-control conditions, visitor access limits and hours, and payment-status outcomes.

4. Results

Test run	Observed result	Coverage shown
Code coverage tests	25 tests passed	100% statement coverage
Branch coverage tests	32 tests passed	100% coverage reported with branch mode enabled

Note: These results reflect the terminal outputs captured during testing. Attach the final code-coverage and branch-coverage PNG screenshots as evidence. Confirm the final report output after any further code changes.

5. Conclusion

The test runs completed successfully: 25 code-coverage tests and 32 branch-coverage tests passed. The recorded coverage reports show 100% coverage for the self_storage.py module. The test suite checks the

main functional paths and boundary conditions for rent calculation, access permission, and payment status. The accompanying screenshots and Git repository provide supporting evidence for the lab submission.

6. Submission Checklist

■ Attach CodeCoverageReport.png.

■ Attach BranchCoverageReport.png.

■ Include CodeCoverageTest_51_self_storage.py and BranchCoverageTest_51_self_storage.py in the assigned repository.

■ Verify that at least two Git commits are present.

Prepared for Lab 6: Code and Branch Coverage Analysis. Verify the final repository state and attach the required screenshots before submission.